

FINANCIAL AI AGENT

An Intelligent RAG-Based Financial Advisor for Local and Global Markets

Document 2: Literature Review & Requirements Gathering

Graduation Project — Digital Egypt Pioneers Initiative (DEPI)
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Part A — Literature Review

A.1 Purpose

This review positions the Financial AI Agent against the landscape of existing conversational finance tools and predictive trading approaches, identifies the specific gaps the project addresses, and records the evaluation process (feedback, improvements, and grading criteria) applied to the work.

A.2 Landscape of Related Approaches

Approach	Strengths	Limitations Relevant to This Project
General-purpose LLM chatbots (no retrieval)	Fluent, flexible conversation; low integration cost	Prone to hallucinating prices/figures; knowledge frozen at training cut-off; no grounding in live markets
Traditional robo-advisors (rules/portfolio-theory based)	Deterministic, auditable allocation logic	Rigid; poor conversational UX; typically global-market focused with weak regional (Egyptian) coverage
Pure quant / algorithmic trading models	Statistically rigorous, backtestable signals	No natural-language interface for retail users; usually single-market, single-asset-class
Retrieval-Augmented Generation (RAG) systems	Grounds LLM output in verified external data; reduces hallucination	Requires additional infrastructure (vector DB, embeddings) and careful retrieval design

A.3 Gap Analysis

- Regional data fragmentation: no single existing platform reliably unifies global equity/FX data with Egyptian-specific instruments such as local gold prices.
- Provider unreliability: major global data providers frequently return missing endpoints (HTTP 404) for specific regional currency pairs, forcing bespoke cross-rate math rather than a direct feed.
- Trust gap in conversational AI: general LLMs confidently invent figures when they lack real-time data, and their static training cut-off makes them blind to current volatility — an unacceptable risk for financial guidance.
- Language accessibility: existing tools are largely English-first, creating a barrier for Arabic-speaking retail investors, and naively mixing Arabic (RTL) with financial tickers (LTR) produces unreadable output.
- Reporting vs. forecasting: most conversational finance tools only report current data; few combine this with a validated predictive layer for future price movement.

The Financial AI Agent addresses each gap directly: a dual-market ingestion pipeline (yfinance + BeautifulSoup scraping), a RAG architecture that mathematically constrains the LLM to retrieved context, an Arabic-first interface with an explicit RTL/LTR formatting fix, and a supervised ML pipeline (documented in financial_advisor.ipynb) that extends the system from reporting into forecasting.

A.4 Feedback & Evaluation

This section is maintained as a live record of supervisor/lecturer feedback throughout the review cycle. It is to be completed by the assigned reviewer at each checkpoint.

Review Date	Feedback / Assessment Summary	Reviewer	Status

A.5 Suggested Improvements

- Expand ingestion pipelines to the Egyptian Stock Exchange (EGX) and cryptocurrency markets.
- Add an automated alerting system (push notifications / email) for AI-monitored price triggers.
- Introduce portfolio-level optimization, allowing users to securely input holdings for AI-driven risk assessment and rebalancing.
- Broaden the predictive model's ticker universe beyond the current 40 large-cap US equities.
- Track additional lecturer/reviewer feedback here as it is received.

A.6 Final Grading Criteria

Component	Weight	Basis
Documentation	25%	Completeness and clarity of the six required deliverable sets (planning, literature review, requirements, design, implementation, final report)
Implementation	35%	Working ingestion pipeline, RAG engine, ML pipeline, and Streamlit application; code quality and structure
Testing	20%	Test case coverage, backtest validation methodology (embargoed cross-validation), bug resolution
Presentation	20%	Final demo, slide deck clarity, and team's ability to explain design decisions

Part B — Requirements Gathering

B.1 Stakeholder Analysis

Stakeholder	Interest / Need	Influence on Requirements
Retail investors (Egypt & global)	Fast, trustworthy answers on prices, currencies, and gold values; investment insight in Arabic or English	Drives accuracy, latency, bilingual UI, and Arabic-first design requirements
Academic supervisors / graders	Evidence of sound engineering practice, correct methodology, and measurable results	Drives documentation depth, backtesting rigor, and KPI transparency
Development team	Maintainable, modular codebase; clear division of responsibilities	Drives architecture choices (RAG separated from ML pipeline), coding standards
Data providers (yfinance, Isagha)	Fair-use API/scraping load, availability	Drives caching, refresh-on-demand design, and error-handling requirements
Future users of expanded scope (EGX, crypto, alerts)	Broader asset coverage and proactive notifications	Informs the non-functional scalability/extensibility requirements

B.2 User Stories & Use Cases

ID	User Story	Acceptance Criteria
US-01	As an investor, I want to ask the price of a global stock or currency in natural language, so that I don't need to check multiple sites.	Returns a current, correctly formatted price within seconds; grounded in retrieved data
US-02	As an Egyptian investor, I want to know today's local gold price, so that I can compare it against global trends.	Gold price reflects the latest scraped Isagha value; refreshed on demand
US-03	As a user, I want to ask follow-up questions that reference earlier parts of the conversation, so that I get deeper analysis without repeating context.	Session state retains prior turns and the model uses them correctly
US-04	As an Arabic-speaking user, I want numbers and tickers to display cleanly inside Arabic text, so the response is readable.	Numeric/ticker values are bracketed and render left-to-right inside RTL paragraphs

ID	User Story	Acceptance Criteria
US-05	As an investor, I want to see which stocks currently show a statistically strong buy signal, so I can prioritize research.	System returns a ranked list of high/moderate-conviction tickers with model probability
US-06	As a user, I want the AI to refuse questions outside its financial scope, so I'm not misled by out-of-context answers.	Boundary-setting prompt causes explicit refusal for non-financial queries
US-07	As a data engineer, I want a manual refresh trigger, so the freshest prices are used before a critical question is answered.	'Update Data' button forces the ingestion pipeline to run before inference

B.3 Functional Requirements

- FR-1: The system shall ingest global equity and currency data via the yfinance API on a scheduled and on-demand basis.
- FR-2: The system shall scrape and clean Egyptian gold prices from the Isagha platform and store them alongside global data.
- FR-3: The system shall generate vector embeddings (BAAI/bge-m3) for ingested data and store them in PostgreSQL with pgvector for semantic retrieval.
- FR-4: The system shall answer user questions via an LLM (Llama-3.3-70b-versatile via Groq) constrained to retrieved, sourced context.
- FR-5: The system shall refuse to answer questions unrelated to finance or outside the scope of available data.
- FR-6: The system shall format numeric values, currencies, and percentages consistently, with explicit bracketing to preserve readability in mixed Arabic/English text.
- FR-7: The system shall provide a conversational Streamlit interface with persisted session history.
- FR-8: The system shall provide a manual 'refresh data' control that reruns ingestion before answering.
- FR-9: The system shall run a supervised ML pipeline that computes technical-indicator features (RSI, MACD, Bollinger Bands, volatility, momentum, sector-relative measures) for a defined ticker universe.
- FR-10: The system shall classify, for each ticker, whether a statistically significant price increase (>5% in 10 trading days, or volatility-adjusted equivalent) is likely, and output a probability score.
- FR-11: The system shall rank and present tickers as 'high conviction' or 'moderate conviction' buy candidates based on calibrated probability thresholds.
- FR-12: The system shall backtest generated trading signals with realistic transaction costs and position-sizing caps, and report Sharpe, Sortino, Calmar, drawdown, and win-rate metrics against a benchmark (SPY).

B.4 Non-Functional Requirements

Category	Requirement	Rationale / Target
Performance	Chat responses generated with near-instantaneous latency	Groq LPU inference of a 70B-parameter model to preserve a real-time conversational feel
Reliability	100% uptime for critical regional currency pairs	Achieved via USD-based cross-rate computation as a fallback to unreliable direct feeds
Data Quality	Ingested OHLC data must satisfy $\text{Low} \leq \text{Open/Close} \leq \text{High}$ and $\text{Volume} \geq 0$	Automated data-quality filtering rejects malformed rows before feature engineering
Security & Privacy	Embeddings computed locally (not via third-party cloud APIs)	Reduces recurring API cost and preserves data sovereignty/privacy
Usability / Accessibility	Arabic-first UI with correct bidirectional text rendering	Numeric values and tickers bracketed to prevent RTL/LTR clashes
Scalability	Architecture must accommodate additional markets (EGX, crypto) without redesign	Modular ingestion layer decoupled from the RAG and ML layers
Maintainability	ML pipeline changes must be evaluated against a fixed noise-tolerance threshold (0.02 AUC) before adoption	Prevents accepting marginal or noisy improvements as genuine gains
Auditability / No Data Leakage	Time-series validation must use an embargo period between training and evaluation folds	20-day embargo (2× the 10-day label horizon) enforced in <code>get_embargoed_folds()</code>